

Marking scheme [MM 60]

1. (a) phase; $x(n)$ & $x(-n)$ has same magnitude, differentiable by phase (3)

(b) $y(n) = x(2n)$ In general time varying

$$x_{\text{shift}}(n) = x(n+1) \xrightarrow{\text{decimate}} y_{\text{shift}}^{(n)} \rightarrow \text{T.I. if } y_{\text{shift}}^{(n)} = y(n+1) \quad (4)$$

Take an example to prove

(c) Powering Gain $(P_G) = R_C/R_S \Rightarrow P_{G1} = \frac{10}{4.8} \approx 2.083$
 $P_{G2} = \frac{50}{4.8} \approx 10.41$ } $\rightarrow$ 5 times improvement (3)

(d) Zero pad by $(N_1 + N_2 + N_3 - 2) \rightarrow \text{DFT} \rightarrow \text{Multiply} \rightarrow \text{IDFT}$ (3)

(e) $y(n) = \tilde{x}(n) * h(n) = \sum_{k=0}^{N-1} h(k) \tilde{x}(n-k) = \sum_{k=0}^{N-1} h(k) x(n-k)_N$
 $\downarrow$
define
 $= x(n) \otimes h(n) \quad (4)$

2. $x_e(n) \xrightarrow{F} X_R(\omega)$
 $-j x_o(n) \xrightarrow{F} X_I(\omega)$
 $x_e(n) = \{\frac{1}{2}, 0, 1, 2, 1, 0, \frac{1}{2}\}$ (2)
 $x_o(n) = \{\frac{1}{2}, 0, -2, 0, 2, 0, -\frac{1}{2}\}$ (2)

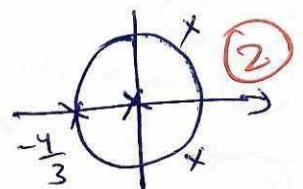
$y(n) = 2\pi \cdot x_e(n) \cdot x_o(n)$ [Multiplication in TD $\leftrightarrow$ conv. in freq. domain]

$y(n) = [\frac{\pi}{2}, 0, -4\pi, 0, 4\pi, 0, -\frac{\pi}{2}]$ (3)

3. From plot $X(z) = \frac{z}{(z+3/4)(z^2-z+1/2)}$; $|z| > 3/4$

$y(n) = x(-n+3) = x(-(n-3)) \Rightarrow Y(z) = z^{-3} X(z^{-1})$

$Y(z) = \frac{8/3}{z(2-2z+z^2)(1/3+z)} \Rightarrow \text{poles} \Rightarrow 0, -\frac{4}{3}, 1 \pm j$
 $\text{zeros} \Rightarrow \infty$



ROC: $0 < |z| < 4/3$ (2)

4. Given $x(0)=1, x(1)=0, x(2)=2, x(3)=2, x(4)=b, x(5)=1$

$$X(e^{j\omega}) = 1 + 2e^{-j2\omega} + 2e^{-j3\omega} + be^{-j4\omega} + e^{-j5\omega}$$

$$x_1(k) = X(e^{j\omega}) \Big|_{\omega=\frac{\pi}{2}k} \quad \text{Cross time domain Aliasing of } x(n)$$

$N=4 \uparrow$

$$x_1(0) = x(0) + x(4) = 1 + b; \text{ Given } x_1(0) = 4 \Rightarrow \boxed{b=3}$$

$$x_1(1) = x(1) + x(5) = 1$$

$$x_1(2) = 2, \quad x_1(3) = 2$$

(7)

Marking scheme

$$(1) \quad x(n) = \frac{1}{4} W_N^* \begin{bmatrix} 6 \\ 8+4j \\ -2 \\ 8-4j \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ -3 \\ 1 \end{bmatrix} \quad (4)$$

$$(2) = \sum x(n) = \boxed{6} \text{ Marks for } \underline{Q3}$$

$$(2) \quad x(k) + x^*(-k) \longleftrightarrow x(n) + x^*(-n)_N$$

$$\Rightarrow [2, 6, 6, 6]$$

$$(2) = 2+6 = \boxed{8} \text{ Marks for } \underline{Q4} \quad (4)$$

$$(2) \text{ mod-3 circ. conv } y = x * h = \text{Then mod-3 or any other method}$$

$$[10, 10, 13]$$

$$(2) = \boxed{10} \text{ marks for } \underline{Q5} \quad (4)$$

Total Marks: 60

Q1: (7)

Q2: (7)

(12) for marking scheme soln

(36)

$$(1) = \boxed{6} \text{ Marks } \underline{Q3}$$

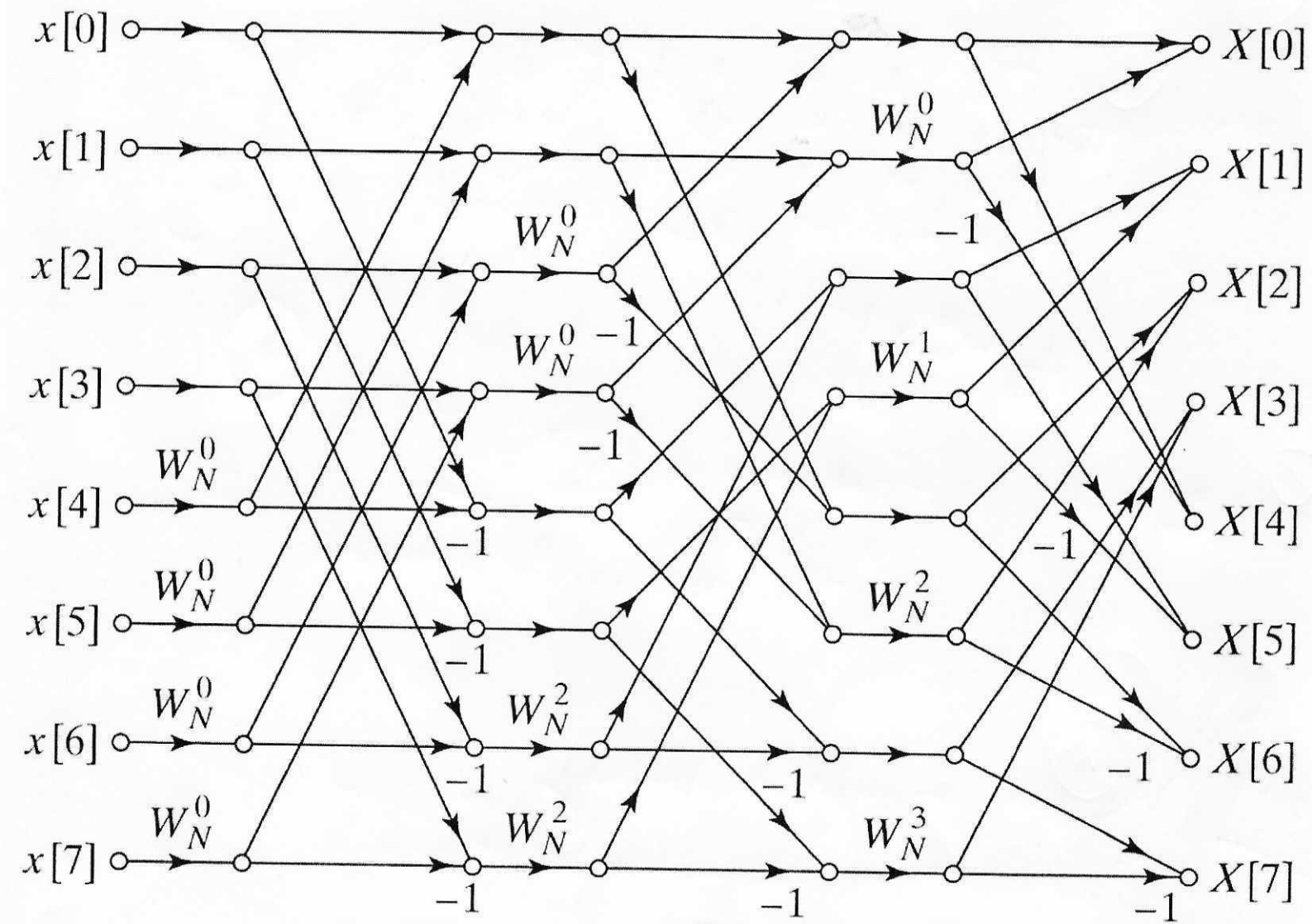
$$(2) = \boxed{8} \text{ Marks } \underline{Q4}$$

$$(2) = \boxed{10} \text{ Marks } \underline{Q5}$$

(24) Marks

Total (60) Marks

5. Rearrangement with both i/p & o/p in normal order



Similar to last stage
of normal DFT
as i/p are in order
now.

10

→ Required explanations to arrive at second &

overhead for index computations / third stages.
highly complicated scheme

→ This scheme requires more memory (RAM), No obvious advantage.